

Distributive Property & Expanding

The distributive property is the engine of algebra. Learn to multiply a bracket by a term, multiply two brackets, and tidy up by combining like terms.

Multiplying out brackets

The rule: Multiplication **distributes** over addition — multiply the outside term by **each** term inside the bracket.

$$a(b + c) = ab + ac$$

Multiplying two brackets

Multiply **every** term in the first bracket by every term in the second, then add the results.

$$(a + m)(b + n) = ab + an + mb + mn$$

This is the master identity of the chapter. With m and n possibly negative, it covers every "expand the brackets" question.

Like terms

After expanding, add together **like terms** — terms with exactly the same letters and powers. $3a^2b + 2a^2b = 5a^2b$, but a^2b and ab^2 cannot be combined.

Why it's useful: the distributive property even gives quick mental-maths tricks. $23 \times 27 + 23 = 23(27 + 1) = 23 \times 28$.

Prerequisite refresher: "Expand" means remove the brackets by multiplying out. We usually drop the $\times$ sign: $a(b+c)$ means $a \times (b + c)$, and $5a$ means $5 \times a$.

Where marks slip away

Trap 1 — Multiply every term. $a(b + c) = ab + ac$, not $ab + c$. The outside term hits both inside terms.

Trap 2 — Signs. $-3p(-5p + 2q) = +15p^2 - 6pq$. A negative times a negative is positive; keep track of every sign.

Trap 3 — Two brackets = four products. $(a + m)(b + n)$ gives four terms before you simplify. Don't multiply only the "firsts" and "lasts".

Trap 4 — Only like terms combine. a^2b and ab^2 are different; they cannot be added into one term.

Slip 5 — Powers when multiplying. $p \times p = p^2$, not $2p$.

Model answers — write them like this

Example A. Expand $3a(a - b + 1)$.

Step 1: Multiply $3a$ by each term inside.

Step 2: $3a \times a = 3a^2$; $3a \times (-b) = -3ab$; $3a \times 1 = 3a$.

$$\therefore 3a^2 - 3ab + 3a$$

Example B. Expand $(3 + u)(v - 3)$.

Step 1: Multiply each term of the first bracket by each term of the second.

Step 2: $3 \times v + 3 \times (-3) + u \times v + u \times (-3) = 3v - 9 + uv - 3u$.

$$\therefore uv - 3u + 3v - 9$$

Example C. Expand $(10a + b)(10c + d)$.

Step 1: Four products: $10a \times 10c + 10a \times d + b \times 10c + b \times d$.

Step 2: $= 100ac + 10ad + 10bc + bd$.

$$\therefore 100ac + 10ad + 10bc + bd$$

Example D. Simplify $3a^2b + 2a^2b + 4ab^2$.

Step 1: $3a^2b$ and $2a^2b$ are like terms: $3a^2b + 2a^2b = 5a^2b$.

Step 2: $4ab^2$ has different letters/powers, so it stays separate.

$$\therefore 5a^2b + 4ab^2$$

Next: [try the worksheet](#), then [check the answer key](#).